MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6DJ8

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: *Double triode, miniature, medium Mu*

Outline	---	6-2 (EIA)
Base	---	E9-1
Envelope	---	T6-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	2a	2g	2k	h	h	1a	1g	1k	int sd

ABSOLUTE-MAXIMUM RATING:

Parameter:	*Ef*	Eb	*Ebb*	*Ec*	*Pp/p*	*Ik/k*	Rg	*Eh1k*	*Eh2k*	*TE*	*Alt*	
Unit:	V	Vdc	Vdc	Vdc	W	mAdc	Meg	v	v	°C	ft	
Maximum:	6.9	130	550	-50	1.8	25	1.0	50	150	130	(see note 2)	Ⓐ
Minimum:	5.7	---	---	---	---	---	---	---	---	---	---	
			(see note 1)						(see note 3)			
TEST CONDITIONS:	6.3	90	---	-1.3	---	---	---	---	---	---	---	

GENERAL:

Qualification - Required

Ⓐ denotes changes

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:58Z
Check the source to verify that this is the current version before use.

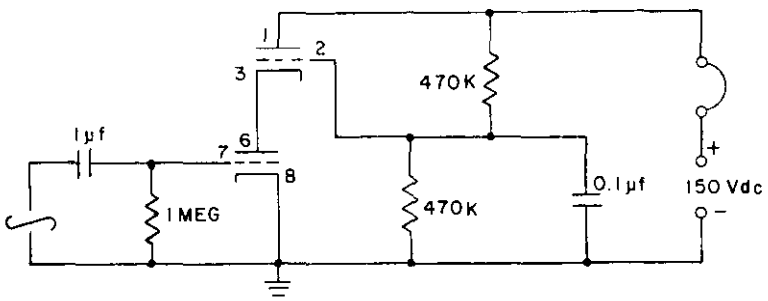
METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1266	Total grid current	See notes 4 and 10	0.65	II	Ic	0	-1.0	μAdc
1306	Transconductance (1)	See note 4	0.65	II	Sm	9,500	16,500	μmhos
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1336	Heater-cathode leakage	See note 4	4.0	II	Ihk	---	20	μAdc
1301	Heater current		4.0	S3	If	330	400	mA
1256	Electrode current (1) (anode)	See note 4	4.0	S3	Ib	9.0	23.0	mAdc
1256	Electrode current (2) (anode)	Ec1 = -5 Vdc (see note 4)	4.0	S3	Ib	---	1.0	mAdc
1306	Transconductance (2)	Ef = 5.7 V (see note 4)	2.5	I	Sm	8,500	---	μmhos
1306	Transconductance (3)	See note 5	4.0	S3	Sm	11,000	---	μmhos
1316	Amplification factor	See note 4	4.0	S3	Mu	29	37	---
1211	Insulation of electrodes		4.0	S3	R	100	---	Meg
1246	Audio frequency noise	Esig = 50 mVac; Ebb = 250 Vdc; Ec1 = -4 Vdc; Rg = 0.1 Meg; Rp = 0.05 Meg (see note 6 and 12)	2.5	I	---	---	---	---
---	Grid-to-cathode breakdown	See notes 4, 8, and 9	4.0	S3	B'dwn	300	---	Vdc
1331	Direct-interelectrode capacitance	No external shield (see note 4)	6.5	Code E	Cgp	1.1	1.7	pF
		No external shield (triode No. 1)			Cin	2.7	4.0	pF
		No external shield (triode No. 2)			Ck-g, h and int sd	5.2	7.0	pF
		No external shield (triode No. 1)			Cout	1.4	2.2	pF
		No external shield (triode No. 2)			Cp-g, h and int sd	2.1	3.5	pF
Ⓐ2126	Glass strain		4.0	I	---	---	---	---
1121	Base strain	See note 7	---	---	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:58Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3							
1501	Intermittent life	Group A; Eb = 125 Vdc; Ec = 0; Rk/k = 180 ohms; Rg/g = 1.0 Meg; Ehk = 150 Vdc with 35 Vac superimposed; (see note 11)	---	---	---	---	---	--- Ⓐ
---	Life-test end points (intermittent) (500 hours)	Total grid current	---	---	Ic1	0	-3.0	μAdc
		Transconductance (1)	---	---	Sm	8,000	---	μmhos

NOTES:

1. Anode voltage at the tube may be permitted to reach this value only under cutoff conditions.
2. See "Reduced pressure (altitude) rating," and altitude, maximum peak voltage. Ⓐ
3. The dc component shall not exceed 130 v.
4. Test each section separately.
5. Test in the cascade connection as shown:

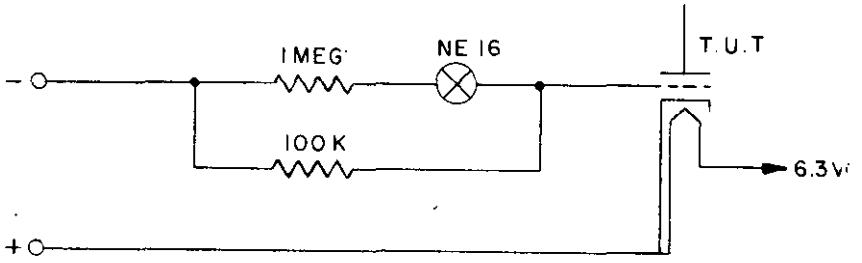


6. Tie 1k to 2k; 1g to 2g; 1a to 2a.
7. Electrical rejects other than inoperatives may be used in the performance of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:58Z
Check the source to verify that this is the current version before use.

NOTES: -Continued

8. In the circuit below increase the dc supply voltage (not exceeding 330 volts) until initial arcing occurs as indicated by the first flickering of the neon lamp.



9. Tubes subjected to grid-to-cathode breakdown and other destructive tests shall not be delivered on the contract or order.

10. This test to be performed at the conclusion of the holding period.

11. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226) will be satisfied if a tube having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack. (A)

12. The rejection level shall be set at the VU meter reading obtained during calibration. (A)

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MI
Navy - SH
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity:
Air Force - 11

(Project 5960-2289-19)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:58Z
Check the source to verify that this is the current version before use.